Spherical hypertopes from marked kernels and antipodal flags

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Abstract

We give a common argument for the spherical classification in the fixed manuscript *An exploration of locally spherical regular hypertopes*. Maxwell's normal-subgroup theorem is retained as a named input. Preservation of the marked Coxeter diagram then leaves only the universal complex and, where it exists, its antipodal quotient. A single explicit flag excludes all even D_n antipodal quotients. For E_7 and E_8, the remaining geometric condition reduces to exact orbits of fundamental weights, avoiding an enumeration of the full groups. Complete orbit certificates accompany the argument.

1 Retained inputs and the marked quotient

We use the following standard facts about a finite Coxeter system (W, S). Its geometric representation is faithful and orthogonal; a closed fundamental chamber meets each orbit in exactly one point. The stabilizer of a face is its indicated standard parabolic subgroup. The spherical Coxeter complex is a flag complex, and W acts regularly on its chambers. The irreducible finite diagrams and their group orders are the usual finite Coxeter classification, also used in Table 1 of the source. These facts are inputs, not new classifications established by this manuscript.

The additional normal-subgroup input is Maxwell's Theorem 0.1, quoted as Theorem 4.1 of the fixed source: if W is a finite irreducible Coxeter group and $N \triangleleft W$, then either $N = \{1, -1\}$ or N is the kernel of a graph homomorphism to a finite Coxeter group. Here a graph homomorphism sends each simple generator to a simple generator or to the identity, and its images contain every simple generator of the target. The full Maxwell paper has not been independently acquired; this is an explicitly retained classification input through the fixed source, with its stated meaning of "graph homomorphism."

For a regular hypertope, the distinguished chamber automorphisms are nonidentity involutions. Their pairwise product orders are exactly the labels of its Coxeter diagram, including order two for an absent edge. The resulting marked group is a quotient of the Coxeter system with that diagram. It is a C-group, and its coset geometry is the original hypertope. These are the acquired reconstruction facts of the L052 general manuscript. In particular, distinct distinguished generators cannot become equal.

Lemma 1. *Let $(W, \{s_i\}_{i\in I})$ and $(W', \{t_j\}_{j\in J})$ be Coxeter systems. Suppose a graph homomorphism $f : W \to W'$ sends every s_i to a nonidentity element, sends distinct s_i to distinct elements, and preserves the order of every product $s_i s_k$. Then f is an isomorphism of marked Coxeter systems, after a bijection of their index sets.*

Proof. Write $f(s_i) = t_{\sigma(i)}$. The assumptions on images make σ injective; the definition of graph homomorphism makes it surjective. The order assumption says that the two Coxeter matrices are equal under σ. The defining presentations therefore give a homomorphism in the opposite direction, taking $t_{\sigma(i)}$ to s_i. Both compositions fix every generator, so both are identity homomorphisms. □

Theorem 2. *Subject to the stated Maxwell input, a regular hypertope whose diagram is finite and irreducible is the universal Coxeter complex or its antipodal quotient. The latter possibility requires $-1 \in W$ and requires the quotient coset geometry to be a hypertope.*

Proof. Let G be the distinguished group of the hypertope. Its marked epimorphism $W \to G$ has kernel N. If Maxwell's alternative expresses N as the kernel of a graph homomorphism $f : W \to W'$, then $W' \cong W/N \cong G$. Thus the simple images of f are nontrivial, distinct, and preserve every original product order. Lemma 1 gives $N = 1$. The remaining alternative is precisely $N = \{1, -1\}$. Reconstruction identifies the actual hypertope with the coset geometry of the indicated quotient and marking. No assumption that its proper residues are universal has been used. □

This argument avoids a separate inspection of the normal-subgroup tables for each finite type. It also avoids using injectivity on a parabolic before knowing that the corresponding residue is universal.

2 Parabolics in the antipodal quotient

Let $\alpha_1, \dots, \alpha_n$ be simple roots in the faithful real representation, and choose fundamental vectors ω_i with $\langle \omega_i, \alpha_j \rangle = \delta_{ij}$. Their Gram matrix is the inverse of the simple-root Gram matrix. For an irreducible finite system,

$$c_{ij} := \langle \omega_i, \omega_j \rangle > 0 \qquad (1 \le i, j \le n). \tag{1}$$

For completeness, write the positive definite simple-root Gram matrix as $D^{1/2}(1 - A)D^{1/2}$, where D is positive diagonal and A has nonnegative entries and connected support. Positive definiteness implies that the largest eigenvalue of A is less than one. For a symmetric nonnegative matrix every eigenvalue has absolute value at most its largest eigenvalue; this follows from $|x^T A x| \le |x|^T A |x|$. Hence $(1 - A)^{-1} = \sum_{m \ge 0} A^m$. Connectivity supplies a positive entry in some power for every pair of indices, proving (1).

Write $W_i = \langle s_j : j \ne i \rangle$. This subgroup fixes the nonzero vector ω_i, so it cannot contain -1. Moreover,

$$-1 \notin W_i W_j \qquad (i \ne j). \tag{2}$$

Indeed, if $-1 = ab$ with $a \in W_i$ and $b \in W_j$, then $a\omega_j = -\omega_j$ and $a\omega_i = \omega_i$. Orthogonality would give $c_{ij} = -c_{ij}$, contrary to (1).

For rank at least three, the quotient by $\{1, -1\}$ is therefore injective on every proper parabolic and satisfies the exact product-kernel criterion proved in *Kernels and intersections in quotients of hypertopes*. It is a C-group on the induced generators. This conclusion does not assert flag transitivity: the latter is precisely what fails in even type D_n. For the string diagrams, the standard string C-group reconstruction theorem does supply a regular polytope and hence a regular hypertope.

3 Incidence and signs

The vertices of type i in the spherical Coxeter complex can be represented by $W\omega_i$. Two vertices of distinct types are incident exactly when they belong to one chamber. We need the following precise numerical test.

Lemma 3. *For $v \in W\omega_j$, one has $\langle \omega_i, v \rangle \le c_{ij}$. Equality holds if and only if $v \in W_i\omega_j$, equivalently if v is incident to ω_i.*

Proof. If v is not in the closed fundamental chamber, some simple root satisfies $\langle v, \alpha_l \rangle < 0$. Reflect in that root. This increases the inner product with $\eta = \sum_k \omega_k$ strictly, and does not decrease the inner product with ω_i: the increase in the latter is zero unless $l = i$, in which case it is strictly positive. The orbit is finite, so repeating reaches its unique dominant representative ω_j. This proves the inequality. If equality held initially, no step with $l = i$ could have occurred.

All steps therefore belong to W_i, and $v \in W_i\omega_j$. Conversely W_i fixes ω_i, so every member of this orbit gives equality. The parabolic description of the Coxeter complex identifies $W_i\omega_j$ as precisely the vertices of type j incident to ω_i. □

Suppose $-1 \in W$. An antipodal vertex is $[v] = \{v, -v\}$. By Lemma 3, two such vertices of distinct types are incident exactly when their chosen representatives have inner product c_{ij} or $-c_{ij}$. Positivity makes the sign unambiguous.

Lemma 4. *Every flag in the antipodal quotient lifts to a flag in the spherical complex if and only if every incident triple has positive product of its three incidence signs. It suffices to test that there are no distinct i, j, k and $v \in W\omega_k$ such that*

$$\langle \omega_i, v \rangle = c_{ik}, \qquad \langle \omega_j, v \rangle = -c_{jk}. \tag{3}$$

Proof. Changing the sign of one representative changes the signs of its incident pairs, and leaves the product around a triangle unchanged. If all triangle products in a flag are positive, choose a first vertex and orient every other representative to have positive inner product with it. The triangle condition makes every remaining pair positive as well. These representatives are pairwise incident in the original complex; its flag property makes them a flag and hence a subflag of a chamber. Conversely any lifted flag admits all-positive representatives.

If some triangle has negative product, orient its first two representatives to make their pair positive. They are incident in the original complex, so an element of W takes them to ω_i, ω_j. Choose the sign of the third representative to make its pairing with ω_i positive. Its pairing with ω_j must then be negative, which is exactly (3). This procedure is reversible. □

When the condition holds, every quotient flag lies in the image of a chamber, and $W/\{1, -1\}$ is flag transitive. The C-group reconstruction with flag transitivity now gives the hypertope. This is stronger than merely checking that every face has the correct number of adjacent chambers.

4 A flag excluding every even D_n quotient

Use the standard realization of D_n by signed permutations with an even number of sign changes, where $n \geq 4$ is even. Its fundamental vertices include

$$e_1, \qquad w = \tfrac{1}{2}(e_1 + \cdots + e_{n-1} - e_n), \qquad v = \tfrac{1}{2}(e_1 + \cdots + e_n).$$

The central antipode belongs to this group exactly when n is even. Put $x = e_n$. The vertices v, w are incident, as they are in the displayed fundamental chamber. The vertices x, v are incident because a coordinate permutation takes e_1 to x and fixes v. Likewise $-x, w$ are incident: apply the permutation interchanging coordinates $1, n$, followed by sign changes in those two coordinates. This group element takes e_1 to $-e_n$ and fixes w.

Consequently $[v], [w], [x]$ form a flag in the antipodal incidence system. Its pairings have signs $+, +, -$, since

$$\langle v, w \rangle = (n-2)/4 > 0, \quad \langle v, x \rangle = 1/2, \quad \langle w, x \rangle = -1/2.$$

It cannot lift to a chamber. This precludes regularity with the proposed marked group: if the quotient were such a regular hypertope, its chambers would be the images of the original chambers by coset reconstruction and chamber transitivity. The negative triple contradicts that requirement. Thus no regular hypertope of type D_n can have the antipodal quotient as its distinguished group. This one flag handles every even rank, without a separate chamber lower bound or induction on rank.

5 The two exceptional incidence calculations

For E_7 and E_8, choose the simply laced Cartan matrix C with diagonal entries two and entries minus one along the tree. Number the trivalent node by 0. Its three arms have vertices (1), $(2,3)$, and $(4,5,6)$ for E_7, with $(4,5,6,7)$ as the third arm for E_8. All other entries are zero. Put $d = \det C$, so $d = 2$ or 1, respectively, and $B = dC^{-1}$. These are integral matrices with strictly positive entries.

Represent $d\omega_k$ in the simple-root basis by column k of B. A simple reflection changes an integer vector z by

$$z \longmapsto z - \left(\sum_l C_{il} z_l \right) e_i. \tag{4}$$

Moreover $\langle \omega_i, z \rangle = z_i$. Therefore the excluded pattern (3) becomes the exact pair of integer equalities

$$z_i = B_{ik}, \qquad z_j = -B_{jk}, \qquad i, j, k \text{ distinct.}$$

Starting at column k, close a set under all the reflections (4). Every stored vector belongs to the orbit, by induction on its construction. When no new vector is found, the set is invariant under every generator and contains the starting vector, so it is the full orbit. Thus no group-order estimate or randomized sampling enters the coverage argument.

The complete orbit certificates have the following sizes and contain no excluded pattern. A separate reader checks that each set contains its initial weight and is closed under every simple reflection. These two conditions already suffice for coverage of the required orbit.

	0	1	2	3	4	5	6	7
E_7	10080	576	2016	126	4032	756	56	–
E_8	483840	17280	69120	2160	241920	60480	6720	240

These fifteen orbits contain 899402 vectors in total. The reader derives B again by rational elimination from the displayed Dynkin trees. It checks strict ordering, uniqueness, the initial weight, every reflection image, and the absence of the excluded coordinate pattern. It does not import the orbit construction. The resulting finite sets are supplied in full, with deterministic compressed copies. The calculation enumerates the fundamental-weight orbits, rather than the groups of orders 2903040 and 696729600.

All arithmetic is integral. Indeed $z^T C z = dB_{kk}$ for an orbit vector; Cauchy–Schwarz gives $z_i^2 \le B_{ii} B_{kk} \le 900$. Every stored coordinate therefore has absolute value at most thirty. The producer checks this bound after each reflection; its intermediate integer expressions are bounded by the displayed matrices. A deadline or storage failure is an incomplete calculation and cannot return a complete certificate. The reader also rejects a set with a deleted vector, a changed coordinate, or an incorrect orbit header; the producer rejects a changed Cartan matrix. These controls exercise coverage and exact input identity.

6 Classification and proper residues

For clarity, the central antipode can be decided independently of the classification under discussion. In type A_n with $n \ge 2$, a coordinate permutation on the sum-zero space cannot negate every difference $e_i - e_j$: three distinct coordinates already give contradictory required images. In B_n every signed permutation occurs, whereas in D_n only an even number of sign changes is allowed. Thus -1 occurs in B_n and exactly the even D_n.

For E_6, the Cartan determinant is three. The weight lattice modulo the root lattice consequently has order three. Each simple reflection acts trivially on this quotient, since its difference from the identity is an integral multiple of a simple root. Negation acts nontrivially

on a group of order three, so it cannot belong to the Weyl group. The determinant and the usual weight/root-lattice identification are retained elementary Coxeter facts here.

For the remaining types, the accompanying exact matrix calculation verifies that the product of simple reflections, raised to the indicated power, is $-I$:

type	E_7	E_8	F_4	H_3	H_4
power	9	15	6	5	15.

The E matrices are those already displayed. For F_4 use the chain Cartan matrix with diagonal two, neighbouring entries minus one except $C_{12} = -2$ (indices start at zero). For H_3, H_4 use the symmetric matrix with diagonal two, first bond $-\phi$, and other chain bonds minus one, where $\phi^2 = \phi + 1$. Reflection i subtracts row i of this matrix from row i of the identity. Multiplication is exact in $\mathbb{Z}[\phi]$, or in $\mathbb{Z}$ in the integral cases. Faithfulness of these retained geometric representations identifies the computed matrix with the antipode in the Coxeter group. In rank one the antipodal quotient loses its generator. In rank two, quotienting $I_2(p)$ by a nontrivial central antipode reduces the order of the product of its generators, and hence changes the prescribed diagram.

The complete finite checks, Theorem 2, the string reconstruction, the even D_n flag, and Lemma 4 give the following classification. The column a records whether the antipodal quotient is also admitted. The group orders in the universal column are retained from the finite Coxeter classification.

diagram	restriction	$\lvert W\rvert$	a
A_n	$n \geq 1$	$(n+1)!$	0
D_n	$n \geq 4$	$2^{n-1}n!$	0
B_n	$n \geq 3$	$2^n n!$	1
$I_2(p)$	$p \geq 3$	$2p$	0
E_6		51840	0
E_7		2903040	1
E_8		696729600	1
F_4		1152	1
H_3		120	1
H_4		14400	1

There are exactly $1+a$ hypertopes in each row. When $a = 1$, the second object has distinguished group of order $|W|/2$. The two objects are distinct because they have different numbers of chambers.

Every proper residue of an admitted antipodal quotient is universal spherical. Indeed a maximal parabolic embeds, as proved above. Flag transitivity and the C-group intersection identities identify the residue of its corresponding face with the coset geometry of that embedded parabolic on its own maximal parabolics. This is its original spherical Coxeter complex. Residues of higher codimension are residues within these. Thus proper spherical residues are a consequence of the classification, not an assumption silently used to force the quotient kernel.

Attribution and limits. The classification conclusions and the standard Coxeter structure belong to the earlier literature. Maxwell's normal-subgroup theorem is a substantial retained input. The work here is the common marked rigidity argument, the explicit antipodal flag criterion, the uniform D_n obstruction, and the exceptional orbit proof with its exact finite obligations. No historical priority or open-conjecture solution is claimed. The Euclidean and hyperbolic classifications in the original manuscript lie outside the scope of this note.

References

[1] M. E. Fernandes, D. Leemans, and A. Ivić Weiss, *An exploration of locally spherical regular hypertopes*. Fixed sixteen-page revised manuscript, catalogue L033; Sections 2–4.

[2] G. Maxwell, *The normal subgroups of finite and affine Coxeter groups*, Proc. London Math. Soc. 76 (1998), 359–382. Theorem 0.1 retained through the fixed source; primary full text not acquired.

[3] Human Compression project, *Residue stabilizers and distinguished generators*; *Kernels and intersections in quotients of hypertopes*. Complete reviewed components acquired with their proofs and stated inputs.